Lab No 7

Docker Setup and PostgreSQL Containers

NAME: LAIBA RANI

REGISTRATION NO.: 2023-BCS-053

COURSE CODE: CS-649

SECTION: A

GITHUB USERNAME: laibarani899-cpu

Task 1: Install Docker and Docker Compose

```
laibarani899-cpu@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
laibarani899-cpu@ubuntu:~$ docker compose version
Docker Compose version v5.6.0
laibarani899-cpu@ubuntu:~$
```

Run both version checks in the same Ubuntu terminal:

Task 2: Run Two PostgreSQL Versions in Docker

```
laibarani899-cpu@ubuntu:~$ docker pull postgres:9.6
9.6: Pulling from library/postgres
1cb79db8a9e7: Pull complete
4b9a74ac6ea0: Pull complete
f6bae7873dd7: Pull complete
f955a6cf127b: Pull complete
f62dc55c568d: Pull complete
11087f2b0d87: Pull complete
8f7722dc50a7: Pull complete
e8622b8cb6f3: Pull complete
d6d74bba3a57: Pull complete
874d4d2a09fd: Pull complete
2d87c3a4038c: Pull complete
1e3d335ef0b7: Pull complete
4e2c4902efbd: Pull complete
01c676df543a: Pull complete
Digest: sha256:caddd35b05cdd56c614ab1f674e63be778e0abdf54e71a7507ff3e28d4902698
Status: Downloaded newer image for postgres:9.6
docker.io/library/postgres:9.6
laibarani899-cpu@ubuntu:~$ docker run postgres:9.6
Error: Database is uninitialized and superuser password is not specified.
       You must specify POSTGRES_PASSWORD to a non-empty value for the
       superuser. For example, "-e POSTGRES_PASSWORD=password" on "docker run".

       You may also use "POSTGRES_HOST_AUTH_METHOD=trust" to allow all
       connections without a password. This is *not* recommended.

       See PostgreSQL documentation about "trust":
       https://www.postgresql.org/docs/current/auth-trust.html
laibarani899-cpu@ubuntu:~$
```

Pull the PostgreSQL 9.6 image.Try to run the image without setting a password:

```
laibarani899-cpu@ubuntu:~$ docker run -d -e POSTGRES_PASSWORD=password postgres:9.6
121aa1d658e50eff763aab5d9cc644b2a2c4808500e493336949eb6a9b4a252c
laibarani899-cpu@ubuntu:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED         STATUS          PORTS
                     NAMES
121aa1d658e5   postgres:9.6                   "docker-entrypoint.s…"   8 seconds ago   Up 7 seconds    5432/tcp
                     ecstatic_grothendieck
0f84cee3db07   sonatype/nexus3:latest         "/opt/sonatype/nexus…"   3 days ago      Up 12 minutes   0.0.0.0:8006->8081/tcp, [::]:
8006->8081/tcp       nexus
5f6b815597b8   intellij-idea-docker-intellij   "/init"                  5 days ago      Up 12 minutes   3000/tcp, 0.0.0.0:3001->3001/
tcp, [::]:3001->3001/tcp   intellij-idea-docker-intellij-1
laibarani899-cpu@ubuntu:~$
```

Start a new PostgreSQL 9.6 container with the password set. List the running containers:

```
laibarani899-cpu@ubuntu:~$ docker run -d -e POSTGRES_PASSWORD=password postgres:10.10
Unable to find image 'postgres:10.10' locally
10.10: Pulling from library/postgres
b18dd0a6efec: Pull complete
80369df48736: Pull complete
c5a7f905c8ec: Pull complete
5c20c5b8227d: Pull complete
5a3f55930dd8: Pull complete
ffc097878b09: Pull complete
69802daaa561: Pull complete
0d24b08575ba: Pull complete
31babd3be108: Pull complete
16f2724dc303: Pull complete
756351b7a443: Pull complete
88d1fc513b8f: Pull complete
f19250dffc5e: Pull complete
3106d02490d4: Pull complete
Digest: sha256:3263c8dad1b5aa6be80144786b18515910165ed771863d686c8b8fda30fea1ad
Status: Downloaded newer image for postgres:10.10
489f067818530c7a7bb6a93a63999c77a96759318469a1957d009b59e50a9904
laibarani899-cpu@ubuntu:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED              STATUS          PORTS
                     NAMES
489f06781853   postgres:10.10                 "docker-entrypoint.s…"   About a minute ago   Up 24 seconds   5432/tcp
                     fervent_tharp
121aa1d658e5   postgres:9.6                   "docker-entrypoint.s…"   7 minutes ago        Up 7 minutes    5432/tcp
                     ecstatic_grothendieck
0f84cee3db07   sonatype/nexus3:latest         "/opt/sonatype/nexus…"   3 days ago           Up 19 minutes   0.0.0.0:8006->8081/tcp,
[::]:8006->8081/tcp  nexus
5f6b815597b8   intellij-idea-docker-intellij   "/init"                  5 days ago           Up 19 minutes   3000/tcp, 0.0.0.0:3001->
3001/tcp, [::]:3001->3001/tcp   intellij-idea-docker-intellij-1
laibarani899-cpu@ubuntu:~$
```

Start a PostgreSQL 10.10 container in the background.List the running containers again

Task 3: Inspect Docker and Manage Containers and Images

```
laibarani899-cpu@ubuntu:~$ docker info
Client:
 Version:    29.1.3
 Context:    default
 Debug Mode: false
 Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  0.30.1
    Path:     /usr/libexec/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version:  v5.6.0
    Path:     /usr/local/lib/docker/cli-plugins/docker-compose
  trust: Manage trust on Docker images (Docker Inc.)
    Version:  29.1.3
    Path:     /usr/libexec/docker/cli-plugins/docker-trust

Server:
 Containers: 5
  Running: 4
  Paused: 0
  Stopped: 1
 Images: 4
 Server Version: 29.1.3
 Storage Driver: overlayfs
  driver-type: io.containerd.snapshotter.v1
 Logging Driver: json-file
 Cgroup Driver: systemd
 Cgroup Version: 2
 Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
 CDI spec directories:
  /etc/cdi
  /var/run/cdi
 Swarm: inactive
 Runtimes: io.containerd.runc.v2 runc
 Default Runtime: runc
 Init Binary: docker-init
 containerd version:
 runc version:
 init version:
 Security Options:
  apparmor
  seccomp
   Profile: builtin
  cgroupns
 Kernel Version: 7.0.0-43-generic
 Operating System: Ubuntu 26.04.1 LTS
 OSType: linux
 Architecture: x86_64
 CPUs: 4
 Total Memory: 5.284GiB
 Name: ubuntu
 ID: e0c39017-8ba3-405a-9a07-52d31cef1164
 Docker Root Dir: /var/lib/docker
 Debug Mode: false
 Experimental: false
```

Display information about the Docker installation:

```
laibarani899-cpu@ubuntu:~$ docker search ubuntu
NAME                      DESCRIPTION                                     STARS     OFFICIAL
ubuntu                    Ubuntu is a Debian-based Linux operating sys…   17878     [OK]
ubuntu/squid              Squid is a caching proxy for the Web. Long-t…   132
ubuntu/nginx              Nginx, a high-performance reverse proxy & we…   143
ubuntu/bind9              BIND 9 is a very flexible, full-featured DNS…   121
ubuntu/kafka              Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper          ZooKeeper maintains configuration informatio…   15
ubuntu/apache2            Apache, a secure & extensible open-source HT…   106
ubuntu/python             A chiselled Ubuntu rock with the Python runt…   34
ubuntu/prometheus         Prometheus is a systems and service monitori…   80
ubuntu/jre                Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet      Chiselled Ubuntu runtime image for ASP.NET a…   28
ubuntu/mysql              MySQL open source fast, stable, multi-thread…   74
ubuntu/dotnet-runtime     Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/postgres           PostgreSQL is an open source object-relation…   46
ubuntu/dotnet-deps        Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/redis              Redis, an open source key-value store. Long-…   26
ubuntu/jdk                Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq           Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/memcached          Memcached, in-memory keyvalue store for smal…   8
ubuntu/grafana            Grafana, a feature rich metrics dashboard & …   13
ubuntu/loki               Grafana Loki, a log aggregation system like …   3
ubuntu/traefik            Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/mimir              Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/php                PHP is a general-purpose scripting language …   10
ubuntu/grafana-agent      Ubuntu ROCK for Grafana Agent, an open-sourc…   0
laibarani899-cpu@ubuntu:~$
```

Search Docker Hub for Ubuntu images.

```
laibarani899-cpu@ubuntu:~$ docker images
                                                                                                    Info →  U  In Use
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
intellij-idea-docker-intellij:latest   f79860ea5fac       11.8GB         3.06GB   U
postgres:10.10                       3263c8dad1b5        310MB          77.7MB   U
postgres:9.6                         caddd35b05cd        294MB          73.6MB   U
sonatype/nexus3:latest               91a9af44af77       1.81GB           809MB   U
laibarani899-cpu@ubuntu:~$ docker run -d ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
96aae9022e0e: Pull complete
517b871c26ea: Pull complete
e211d985152c: Download complete
Digest: sha256:065aa8626841f16057b78b027b464cd59c4fac36c204c72dcbf48d66f5fa8a96
Status: Downloaded newer image for ubuntu:latest
39b560fd3e3e69dee9f2d1e9515d26071f397f06151eb678302450b6dd16f7bc
laibarani899-cpu@ubuntu:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED          STATUS          PORTS                                                                  NAMES
489f06781853   postgres:10.10                       "docker-entrypoint.s…"   9 minutes ago    Up 8 minutes    5432/tcp                                                               fervent_tharp
121aa1d658e5   postgres:9.6                         "docker-entrypoint.s…"   14 minutes ago   Up 14 minutes   5432/tcp                                                               ecstatic_grothendieck
0f84cee3db07   sonatype/nexus3:latest               "/opt/sonatype/nexus…"   3 days ago       Up 26 minutes   0.0.0.0:8006->8081/tcp, [::]:8006->8081/tcp                            nexus
5f6b015597b8   intellij-idea-docker-intellij        "/init"                  5 days ago       Up 26 minutes   3000/tcp, 0.0.0.0:3001->3001/tcp, [::]:3001->3001/tcp                  intellij-idea-docker-intellij-1
laibarani899-cpu@ubuntu:~$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED          STATUS                      PORTS                                                                  NAMES
39b560fd3e3e   ubuntu                               "/bin/bash"              2 minutes ago    Exited (0) About a minute ago                                                              competent_dhawan
489f06781853   postgres:10.10                       "docker-entrypoint.s…"   10 minutes ago   Up 9 minutes                5432/tcp                                                               fervent_tharp
121aa1d658e5   postgres:9.6                         "docker-entrypoint.s…"   16 minutes ago   Up 16 minutes               5432/tcp                                                               ecstatic_grothendieck
3a303d41e12d   postgres:9.6                         "docker-entrypoint.s…"   18 minutes ago   Exited (1) 18 minutes ago                                                                  serene_brahmagupta
0f84cee3db07   sonatype/nexus3:latest               "/opt/sonatype/nexus…"   3 days ago       Up 28 minutes               0.0.0.0:8006->8081/tcp, [::]:8006->8081/tcp                            nexus
5f6b015597b8   intellij-idea-docker-intellij        "/init"                  5 days ago       Up 28 minutes               3000/tcp, 0.0.0.0:3001->3001/tcp, [::]:3001->3001/tcp                  intellij-idea-docker-intellij-1
laibarani899-cpu@ubuntu:~$
```

List local images, start an Ubuntu container, and compare the two container lists. Run the four commands in order. Give the new Ubuntu container a moment to exit before running `docker ps`.

```
laibarani899-cpu@ubuntu:~$ docker restart ecstatic_grothendieck
ecstatic_grothendieck
laibarani899-cpu@ubuntu:~$ docker stop ecstatic_grothendieck
ecstatic_grothendieck
laibarani899-cpu@ubuntu:~$ docker rm ecstatic_grothendieck
ecstatic_grothendieck
laibarani899-cpu@ubuntu:~$ docker rmi postgres:9.6
Error response from daemon: conflict: unable to delete postgres:9.6 (must be forced) - container 3a303d41e12d is using its referenced image caddd35b05cd
laibarani899-cpu@ubuntu:~$ docker rmi ecstatic_grothendieck
Error response from daemon: No such image: ecstatic_grothendieck:latest
laibarani899-cpu@ubuntu:~$ docker rmi -f postgres:9.6
Untagged: postgres:9.6
laibarani899-cpu@ubuntu:~$
```

Restart, stop and remove the PostgreSQL 9.6 container, then remove its image tag.